AES Benchmarking on Raspberry Pi 5

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Project Details

Our project's goal was to analyze how architectural implementation choices affect encryption performance on the Raspberry Pi 5. We benchmarked hardware-accelerated AES on the Raspberry Pi 5 using perf paired with OpenSSL and cryptsetup. We used these programs in order to measure throughput, CPU cycles, instructions executed, instructions per cycle, cache miss rate, and temperature of the system.

Background

AES is a symmetric-key encryption algorithm used to secure digital information through encrypting readable plaintext into unreadable ciphertext, and decrypting it back using the same secret key. This transformation is done through repeated rounds of substitution, shifting, mixing, and key-based transformations. Different key lengths of of 128, 192, or 256 bits are used with longer key lengths ensuring stronger security. It is used in applications such as online banking, messaging apps, and protecting sensitive files.

Results

cryptsetup benchmark

CBC Mode (Encrypt / Decrypt)
128-bit: 1141 / 2264 MiB/s
192-bit: 1054 / 2042 MiB/s
256-bit: 947 / 1832 MiB/s

CTR Mode (Encrypt / Decrypt)
128-bit: 2170 / 2179 MiB/s
192-bit: 1896 / 1897 MiB/s
256-bit: 1717 / 1712 MiB/s

CPI (Cycles/Instruction)
CBC: 0.65–0.66 | CTR: 0.46–0.48

Cache Miss Rate
CBC: ~2.6% | CTR: ~2.9%

perf stat + OpenSSL

CBC Mode (CPI)
128: 0.50 | 192: 0.48 | 256: 0.48

CTR Mode (CPI)
128: 0.33 | 192: 0.33 | 256: 0.34

GCM Mode (CPI) — auth overhead
128: 0.36 | 192: 0.35 | 256: 0.36

Cache Miss Rate (all modes)
≈0.00% — near-perfect L1/L2 hit rate

Branch Miss Rate
CBC/CTR: ≈0.01% | GCM: 0.00–0.28%

Importance

AES has a significant impact in preventing sensitive information from being stolen or hacked. This project furthered our knowledge in AES and cybersecurity.

Takeaways

We found that the CTR cipher mode is the fastest, CTR is the fastest mode, usually above 2 GiB/s using the 128-bit key size. On the other hand, the CBC decrypt is ~2x encrypt due to hardware parallelism. Another result we observed is that using OpenSSL and perf shows near-zero cache misses where on the other hand, cryptsetup has around 2.6–2.9% since longer OpenSSL test windows warm caches fully. Lastly, we found that the GCM CPI, usually between 0.35–0.36 falls between CBC and CTR since GHASH authentication adds instructions but stays highly efficient